

Technology Stack

Date	19 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, JavaScript	<i>Used to build a clean, responsive presentation layer featuring a Dashboard Home Page, an intuitive Weather Parameters Input Form, and dynamic tabular templates to render prediction history and real-time results instantly in the browser.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python & Flask Framework	Flask provides a lightweight, flexible WSGI micro-framework ideal for Python-based web applications. It seamlessly handles URL route mapping, executes incoming form parameter validations, loads the saved serialization components via joblib, and passes processed arrays into the machine learning classification layer.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	Serialized File Stores (.pkl / .save) & Local Python logging	Python's native joblib/pickle utility handles file-based persistence for the high-accuracy trained XGBoost model and its associated preprocessing scaler object. Operational histories and transactional input logs are held in a localized relational table format for quick operational lookups.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	IBM Cloud	<i>Chosen as the primary PaaS production hosting environment to deploy the live application. It provides highly scalable cloud server instances, robust infrastructure security, reliable runtime environments for Python Flask stacks, and a publicly accessible URL for global project access.</i>
5	Version Control & CI/CD	Git & GitHub	Git handles local codebase tracking, branch version management, and code checkpoint rollbacks. GitHub acts as the cloud-hosted central repository, enabling secure code backups via standard git

			push routines and establishing a seamless deployment pipeline into the target cloud runtime.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-Learn, XGBoost, and Kaggle API	Open-source Kaggle APIs streamline the download of historical environmental datasets. Scikit-Learn supplies standard feature preprocessing scaling tools, while the XGBoost classification library functions as the high-accuracy analytical algorithm engine that generates real-time probabilistic flood predictions.